AI School #3

Getting convoluted by Convolutional Neural Network 💋

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convoluted

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Agenda

1. **Recap: Neural Networks**
   Finishing up previous work
2. **What's CNN?**
   Convolutional Neural Network?
3. **Going deeper**
   Hyperparameter tuning and transfer learning
4. **CNN Colab**
   Let's dive into code
5. **Workshop Project**
   Neural Style Transfer

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Calling all AI and LLM practitioners!

AGENTIAL

The encyclopedia of popular LLM-based agents prompting methods. We test

Join our team!

By joining us as an open-source contributor, you will have the opportunity to:

- Collaborate on a cutting-edge open-source project
- Sharpen your software engineering skills
- Deepen your understanding of LLMs and LLM-based Agents
- Enhance your research skills through hands-on experience and exploration

What I'm looking for:

- **Strong passion for open-source development**
- **Strong passion for research engineering**
- **An initiative taker and problem solver**
- **proficient experience with Python**

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What is Convolution?

Convolutional Neural Network

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What is Convolution?

You look at a thing with multiple layers at multiple levels.

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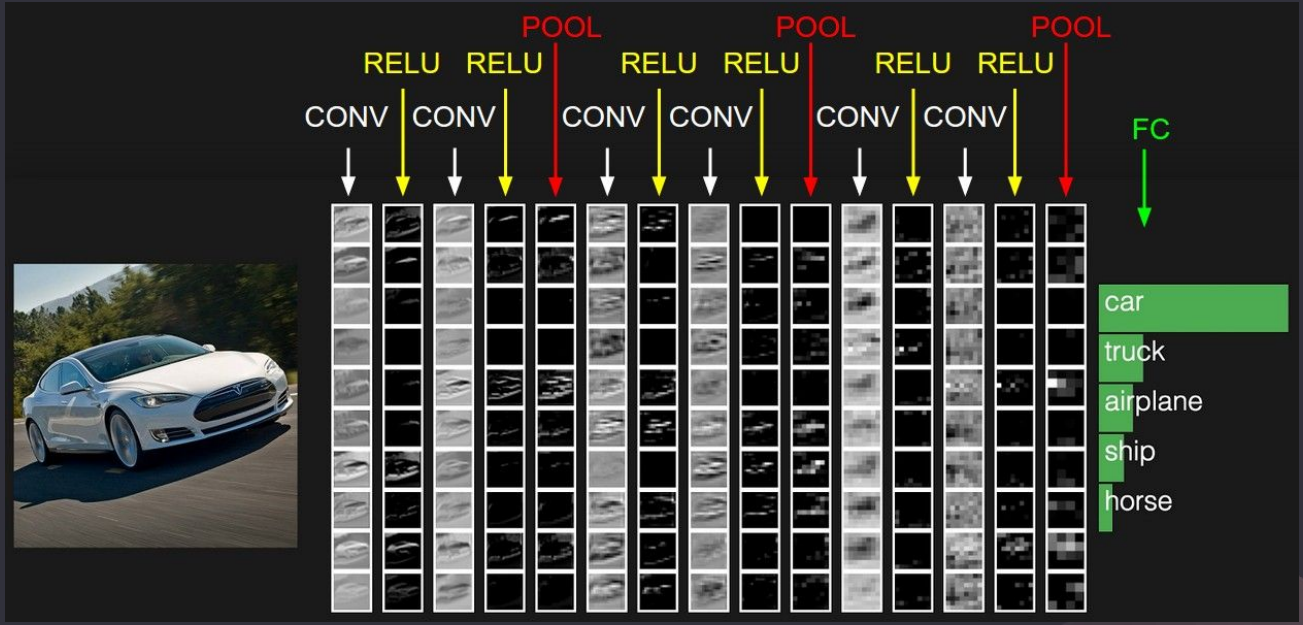
What is Convolution?



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What is Convolution?

- Pooling
- Use of activation function
- Convolutional Layer

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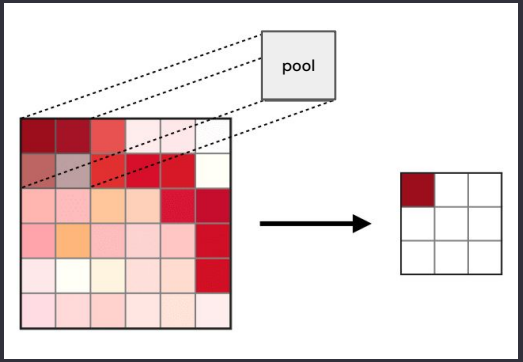
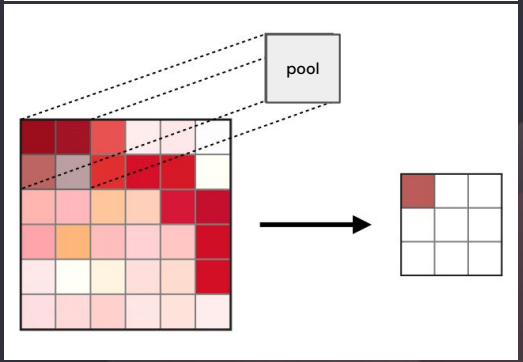
Pooling

Get an image's dominant feature while maintaining its overall picture.

- Reduces overfitting 👁️
- Less data to process ⏰
- Less data less money spent 🤑





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Pooling

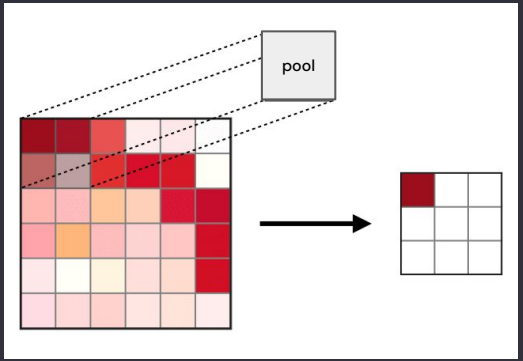
Max Pooling:

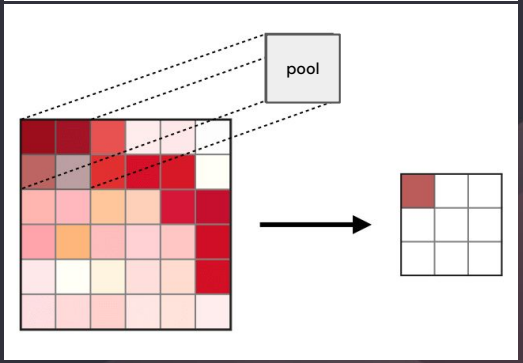
➡️ Takes max value from a set of pixels

Average Pooling:

➡️ Averages a set of pixels.

Which one is max pooling and which one is average pooling?





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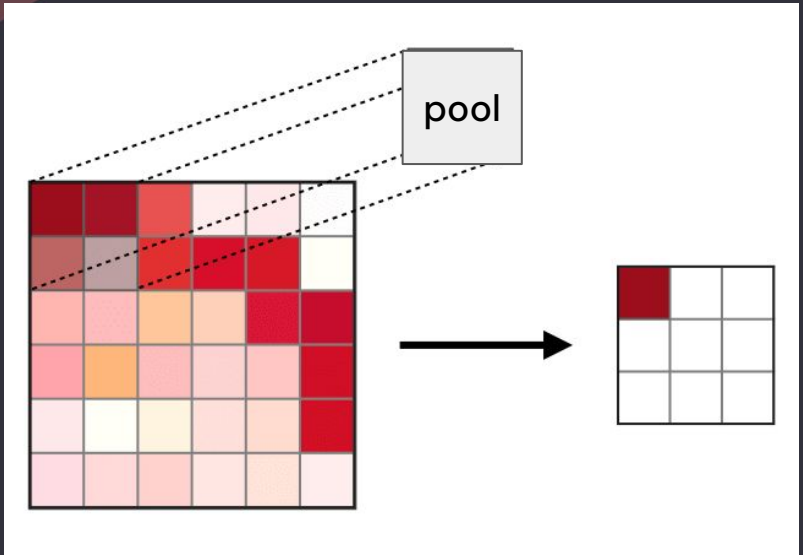
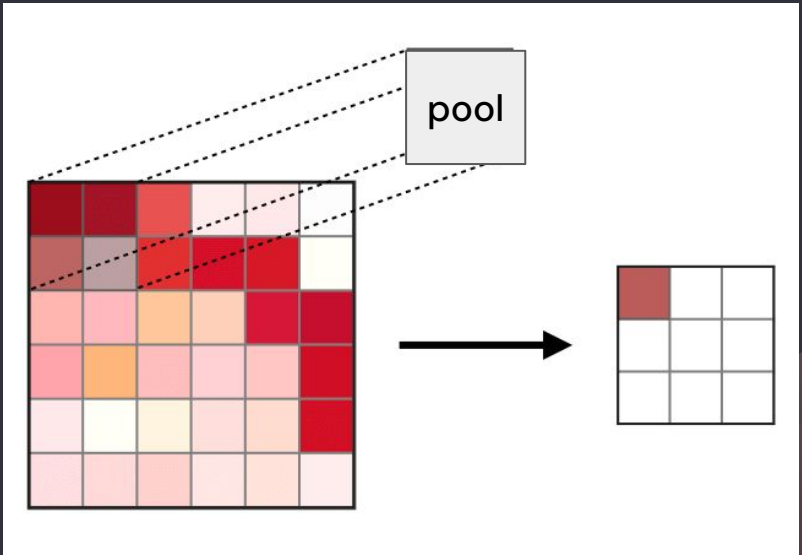
Which one is max pooling and which one is average pooling?





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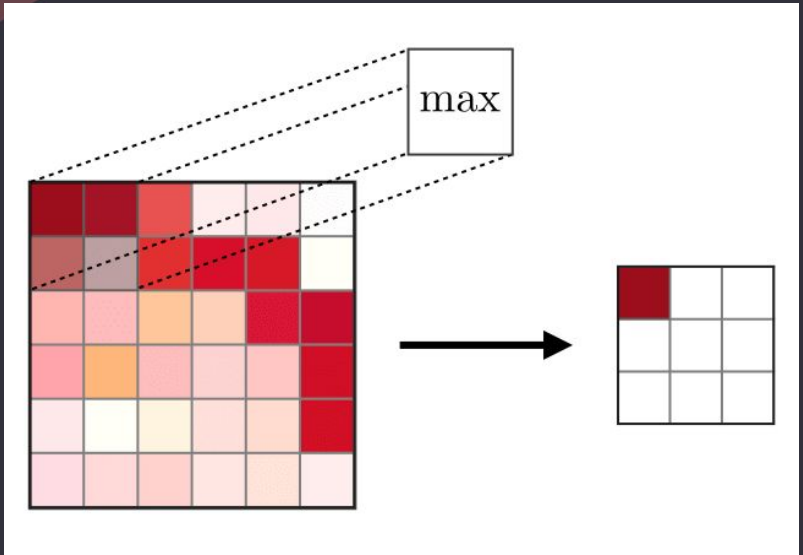
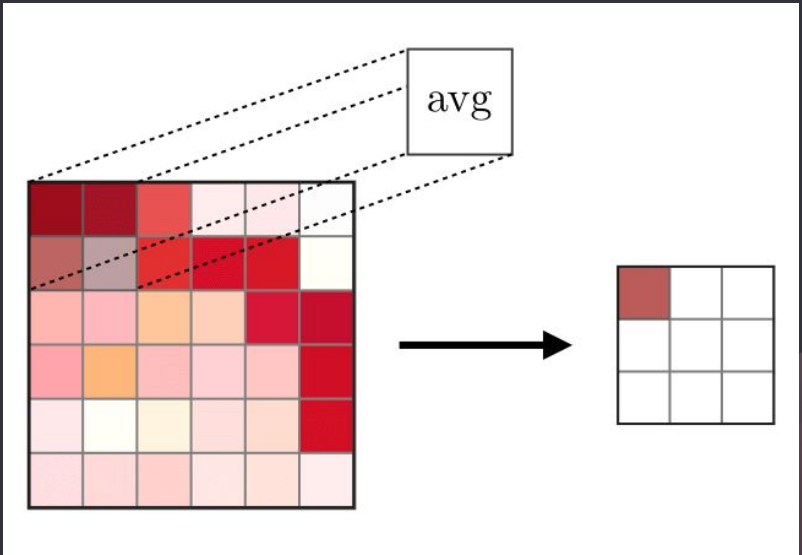
Which one is max pooling and which one is average pooling?





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Do we need this much quality to know its a cat?

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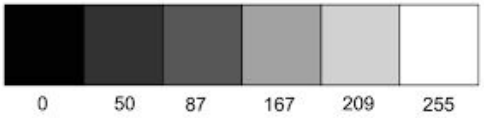

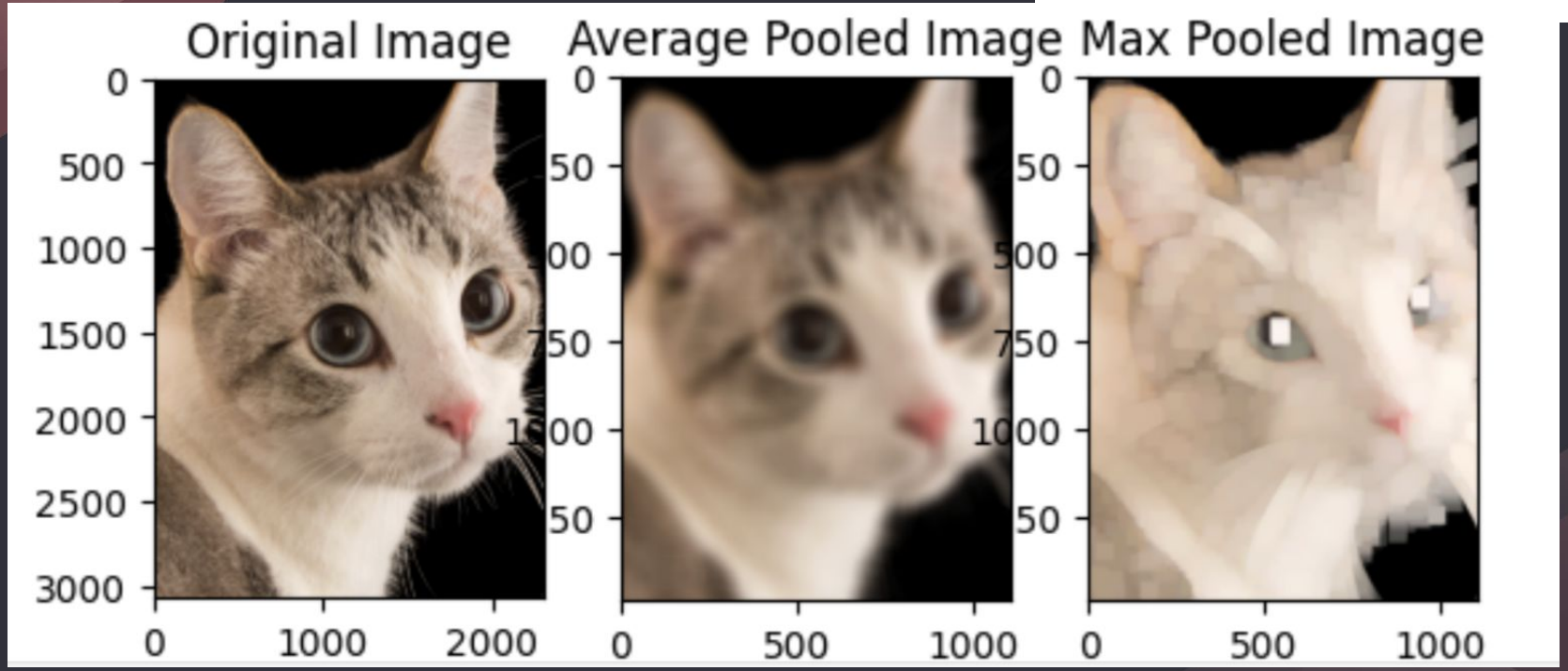

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(Recap) Activation Function

- **Definition**: Activation functions introduce non-linearity into the model, allowing models (CNN in this case) to learn complex patterns.
- **Common Activation Functions**:
  - **ReLU (Rectified Linear Unit)**: Outputs the input directly if it is positive; otherwise, it outputs zero. ReLU is widely used due to its simplicity and ability to mitigate the vanishing gradient problem.
  - **Sigmoid**: Maps input values to a range between 0 and 1, often used in binary classification tasks.
  - **Tanh**: Maps inputs to a range between -1 and 1, providing a stronger gradient than Sigmoid but less commonly used in CNNs.
  - **Softmax**: Converts the output of the network into a probability distribution, where the sum of all probabilities equals 1. It is typically used in the final layer for multi-class classification tasks to determine the class with the highest probability.
- **Role**: Helps in deciding whether a neuron should be activated or not by adding non-linear properties, enabling the network to learn complex patterns.

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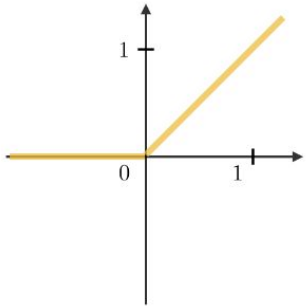
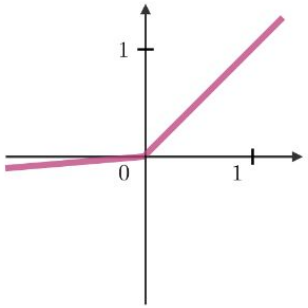
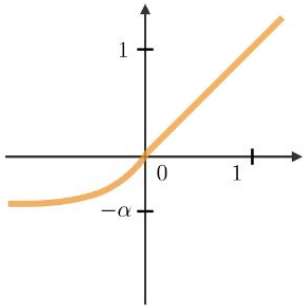
(Recap) Activation Function

ReLU	Leaky ReLU	ELU
$g(z) = \max(0, z)$	$g(z) = \max(\epsilon z, z)$ with $\epsilon \ll 1$	$g(z) = \max(\alpha(e^z - 1), z)$ with $\alpha \ll 1$
		
• Non-linearity complexities biologically interpretable	• Addresses dying ReLU issue for negative values	• Differentiable everywhere

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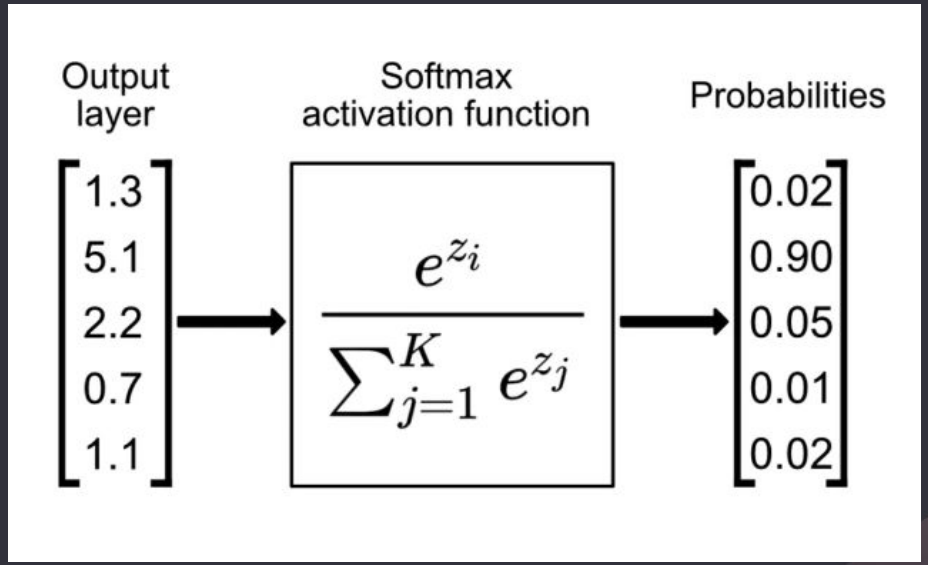
(Recap) Activation Function



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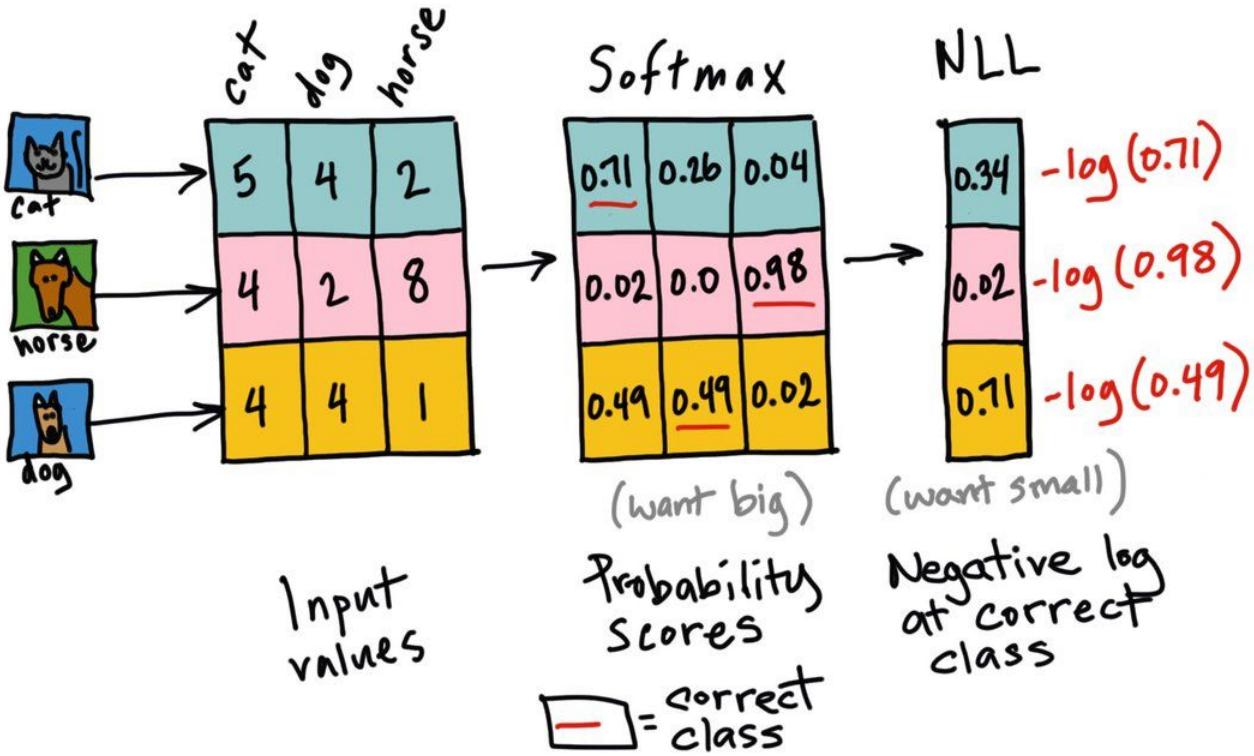
Negative Log Likelihood (NLL) Loss



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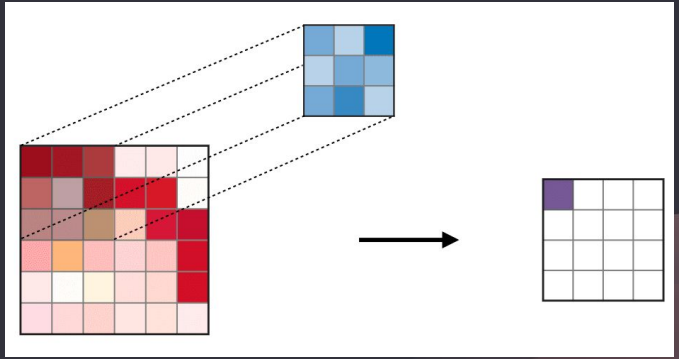
Convolutional Layer

A filter (or kernel) slides over the input image to produce feature maps, capturing local patterns such as edges, textures, and shapes.



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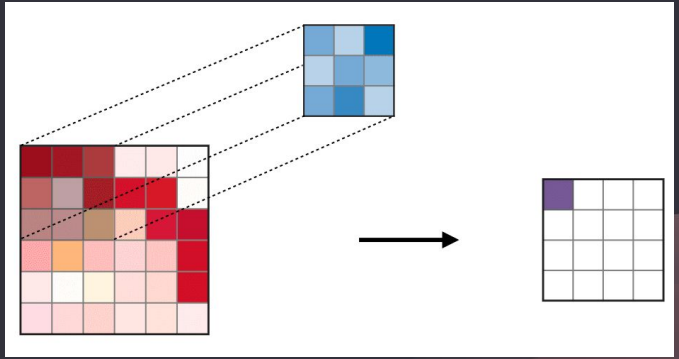
Common Hyperparameters

Kernel Size: Filter Size (e.g., 3x3, 5x5).

Stride: How much the filter shifts over the input image.

Padding: Adds extra pixels around the input image to maintain the size of the output feature map.



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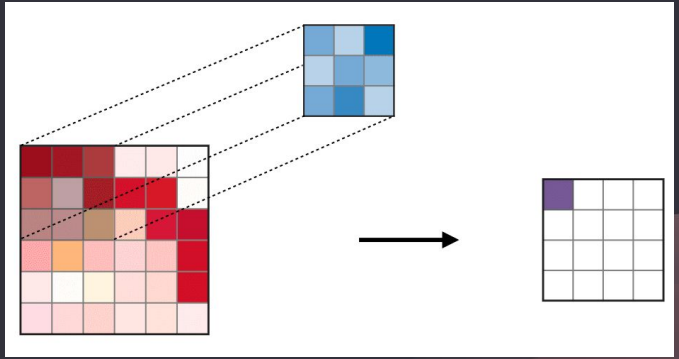
Convolutional Layer

Detects different features at different layers

- lower layers focus on basic features like edges
- deeper layers detect complex structures.



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Goal Comfort your friend by analyzing everything they said about their ex. Apply a series of filters to focus on the negative aspects, so you can reassure them that their ex is the worst

You're given

Everything
their ex did.

Break down everything their ex did and iteratively generalize them to give words of comfort.

Your friend's reaction determines if what you said comforte

If they look comforted, then your words of comfort wo

If they don't, then your words of comfort fails.

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Training Objective

our friend by analyzing everything they said about their ex. Apply a series of filters to focus on the negative aspects, so you can reassure them that their ex is the worst

Input Dataset

Everything their ex did.

Output

Come up with Words of Comfort

Evaluator

If they look comforted, then your words of comfort works.
If they don't, then your words of comfort fails.

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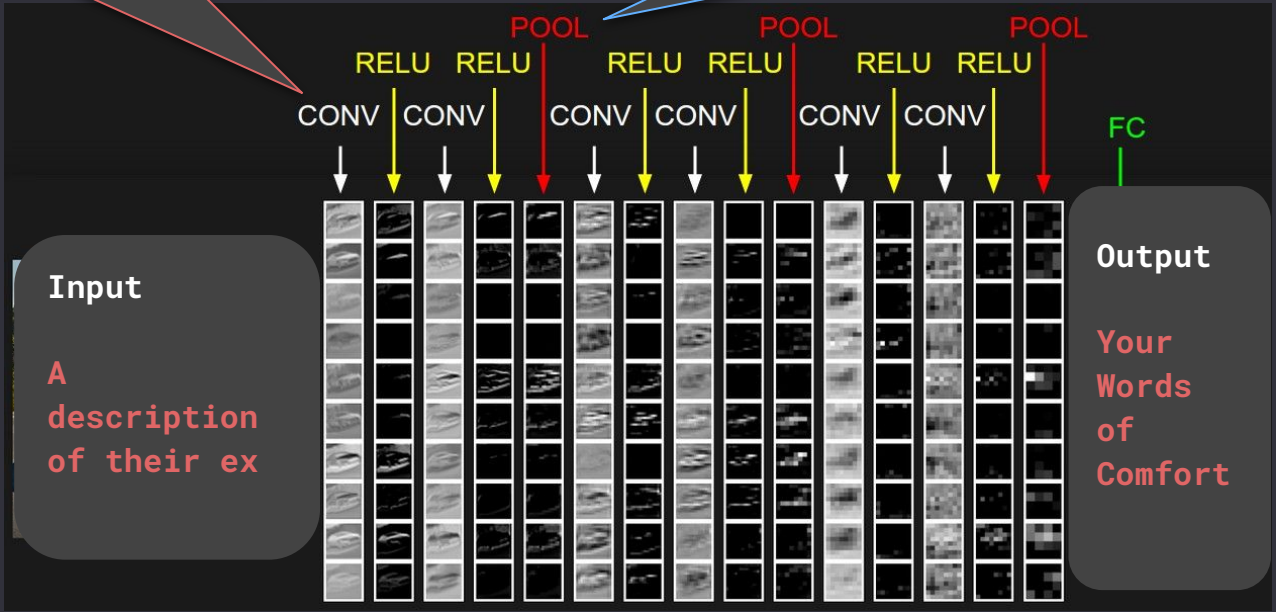
CONV
e.g., "They were always late" → "They were inconsiderate").

POOL
("It sounds like they weren't very respectful of your time.").



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Glossaries 🐈

Input Dataset: "Everything their ex did" is analogous to the raw input data that needs processing in CNN. In a real CNN, this could be image pixels or text data.

Training Objective/Goal: Comfort your friend by filtering out the worst parts of the story about their ex. In CNN terms, this can be thought of as feature extraction, where the model is trained to emphasize negative aspects while downplaying neutral ones.

Filters (Convolutional Layers): CNNs use convolutional filters to detect specific features in data, much like you'd focus on the negative aspects of your friend's complaints to provide comfort. Each layer of filters further refines the details you're focusing on (e.g., "They were always late" → "They were inconsiderate").

Pooling Layers: In CNNs, pooling reduces the size of the data to focus on the most important features. In this analogy, you'd pool or summarize the most crucial complaints ("It sounds like they weren't very respectful of your time.").

Output: "Your words of comfort" is the final output generated by the CNN after processing the input. In this case, the CNN generates a response that makes the friend feel better by emphasizing the negative aspects of their ex. For CNNs, the output might be the classification or the prediction based on input data (e.g., classifying an image as containing a dog or a cat).

Evaluator: The reaction of your friend acts as a loss function in this analogy. If they feel comforted, the model has minimized its loss and achieved its objective; if not, the loss is high, and the model (or you) need to adjust the filters.

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For those curious minded 🐱

(Link) Blog | Stanford's Goated Blog on CNNs

(Link) Interactive | CNN Playground

(Link) Blog | Stanford's CNN Cheatsheet

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Hyperparameter Tuning 🧵

- Manual Hyperparameter Optimization
  (select the parameter with your intuition)
- Grid Search
  (Uniformly sample the values and try different combinations)
- Random Search
  (Get random values and pairing of the hyperparameter values)
- Coarse-to-fine Search
- Bayesian Hyperparameter Optimization

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For those curious minded 😺

(Link) Blog | Google’s Deep Learning Tuning Playbook

(Link) Blog | Andrej Karpathy’s Recipe for Hyperparameter Tuning

(Link) Course Note | Full Stack Deep Learning Note

(Link) Code | Bayesian Optimization Package

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Transfer Learning | Why?

Training takes (lots of) time and money

- **Reduced Training Time**: Use pre-trained models instead of training from scratch.
- **Better Performance**: Often results in higher accuracy for tasks with limited data.
- **Low Data Requirement**: Beneficial when there's insufficient data for a new task.

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Transfer Learning Step 1

- **Pre-trained Model**: Start with a model trained on a large dataset (e.g., ImageNet for computer vision tasks) and finetune it.
- **Feature Extractor**: Use ConvNets pre-trained on ImageNets as image feature extractor. This is really common in modern use of models. E.g., ResNet-50 was used in CLIP, which was used in Meta's open-sourced Vision Language Model Llama 3!

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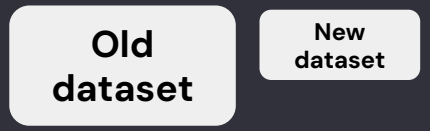
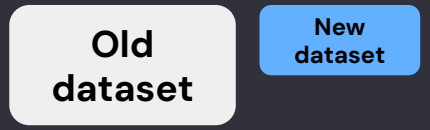
When to finetune?

New dataset is small and similar to original dataset: Avoid fine-tuning to prevent overfitting. Use a linear classifier on CNN codes (higher-level features are still relevant).

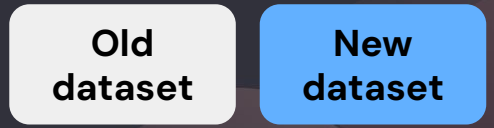


New dataset is large and similar to original dataset: Fine-tuning the full network is safe due to a large dataset reducing overfitting concerns.

New dataset is small and very different from original dataset: Avoid fine-tuning and use a linear classifier on earlier network activations, as the later layers contain more dataset-specific features.



New dataset is large and very different from original dataset: Fine-tune the entire network or train from scratch, but initializing with pretrained weights often provides better results.



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For those curious minded 🐱

(Link) Course Note | Transfer Learning Guide from Stanford CS231n

(Link) Blog | Andrej Karpathy's Recipe for Hyperparameter Tuning

(Link) Course Note | Full Stack Deep Learning Note

(Link) Code | Bayesian Optimization Package

(Link) Slides | Towards understanding transfer learning and its Limits

(Link) Tutorial | Hyperparameter Tuning with PyTorch and Ray Tune

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Colab time

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Let's implement CNNs

Colab link:

acmurl.com/aischool3-colab

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Neural Style Transfer

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What we're going to do

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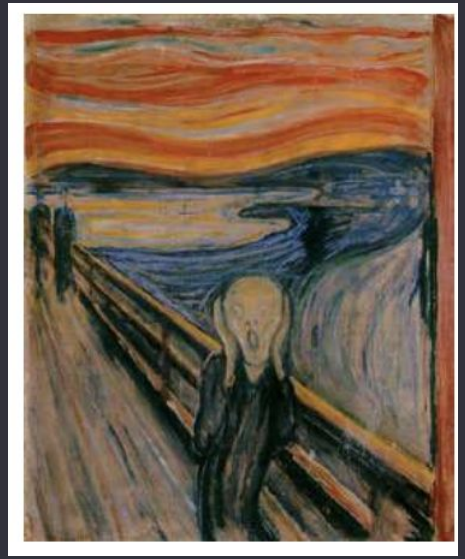

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What we're going to do

How do we do this?

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Select a Pre-trained CNN for Feature Extraction

NST leverages a pre-trained CNN, typically VGG-19 or a similar deep network, which is known to capture robust image features across different layers.

Lower layers capture finer details and textures, while higher layers capture larger structures and content.

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Define Three Images

Content Image: This image provides the main objects and layout (e.g., a landscape or a person).

Style Image: This image provides the artistic style (e.g., brush strokes, colors, patterns).

Generated Image: Initially, this is either random noise or a copy of the content image. This image will be iteratively updated to resemble the style of the style image and the structure of the content image.

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Extract Features from Content and Style Images

Pass the content and style images through the frozen pre-trained CNN to get their feature representations.

For **content** extraction, we use higher layers of the network (e.g., a mid-to-high convolutional layer) that capture the content's structural information.

For **style** extraction, we use multiple layers (both low-level and high-level) because style includes both fine textures and large patterns.

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Define Content and Style Loss Functions

Content Loss: Measures the difference between the features of the content image and the generated image in a specific layer. Commonly, a high layer is chosen for this since it captures the layout and structure without too much detail.
Style Loss: Measures how different the style of the generated image is from the style image.

Style is represented by **Gram matrices** of the features at various layers. A Gram matrix captures the correlations between feature maps, essentially encoding textures and patterns.

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Defining Style and Content Loss

Content Loss =

(MSE)(Content Features,Generated Image Features)

Style Loss =

∑MSE(Gram Matrix of Style Features,Gram Matrix of Generated Image Features)

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Define Total Loss and Optimization

The **Total Loss** is a weighted combination of the content and style losses: Total Loss = α×Content Loss + β×Style Loss

α and **β** are weights that balance the contribution of content and style. Higher α retains more of the content structure, while higher β emphasizes the style.

The objective of NST is to minimize this total loss by adjusting only the pixels in the generated image.

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Optimize the Pixels of the Generated Image

Instead of updating the CNN's parameters, we update the pixels in the generated image to reduce the total loss.

This optimization is done using backpropagation, iteratively adjusting the generated image to match the style and content objectives.

Over many iterations, the generated image takes on the content structure of the content image and the stylistic patterns of the style image.

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Colab time

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Let's implement CNNs

Colab link:

acmurl.com/aischool3-colab

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Thank You

Do you have any questions?

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